Continuum theory. Applying Itô's formula to the empirical number density $\rho(\mathbf{r},t)$ gives a diffusive drift and a Brownian current in the point-particle approximation (see Supplementary Material). To coarse-grain the diffusivity switch, we write the sensed signal as a slow field P plus a fast fluctuation ζ of probability density p_ζ. Holding p_ζ fixed under weak modulations of P, we average the switch to obtain

$$D_{\rm eff}(P) = D_m + (D_M - D_m)\,\Pr(\zeta > P_{\rm th} - P), \qquad (1)$$

where Pr denotes probability over ζ. The resulting Dean-type equation is

$$\begin{aligned} \partial_t \rho &= \nabla^2[\rho D_{\rm eff}(P)] - \nabla\cdot\left[\sqrt{2\rho D_{\rm eff}(P)}\,\mathbf{\Lambda}\right], \\ P &= K_{d_c} * \rho, \end{aligned} \qquad (2)$$

Here $\mathbf{\Lambda}$ is zero-mean Gaussian vector white noise of unit covariance, and $*$ denotes spatial convolution. The kernel is $K_{d_c}(r) = 1/(2\pi r)$ for $0 < r < d_c$ and zero outside, where r is the separation. Both terms are divergences and conserve particle number.

Relaxation and static fluctuations. We linearize Eq. (2) about the homogeneous state, where the mean sensed signal is $\overline{P} = \rho_0 d_c$. Let $\overline{D} = D_{\rm eff}(\overline{P})$ and $g = D'_{\rm eff}(\overline{P})$, with the prime denoting differentiation with respect to P. The relaxation rate of a Fourier mode at q is

$$\lambda(q) = q^2[\overline{D} + \rho_0 g \widehat{K}_{d_c}(q)], \qquad (3)$$

where $\widehat{K}_{d_c}$ is the Fourier transform of the sensing kernel. Ordinary diffusion and sensing both contribute to this rate. The conserved noise supplies power $2\rho_0\overline{D}q^2$. For a stable nonzero mode ($\lambda(q) > 0$), the scalar Lyapunov equation gives the stationary variance. Dividing by ρ_0 gives the total static structure factor

$$S_{\rm tot}(q) = \frac{\overline{D}}{\overline{D} + \rho_0 g \widehat{K}_{d_c}(q)}. \qquad (4)$$

The constant-diffusivity limit $g = 0$ recovers $S_{\rm tot}(q) = 1$ for independent Brownian particles.

Linear spectrum at threshold matching. At $d_c = L_p$, $\overline{P} = P_{\rm th}$, and Eq. (1) gives $g = (D_M - D_m)p_\zeta(0)$. A narrow distribution peaked at zero can give a steep positive response. For $qL_p \gg 1$, the truncated $1/r$ kernel has $\widehat{K}_{d_c}(q) \sim 1/q$. Where sensing dominates diffusion, the relaxation rate is then proportional to q, while the noise power remains proportional to q^2. This balance predicts a linear regime of the form observed in Fig. 2,

$$S_{\rm tot}(q) \sim \frac{\overline{D}}{\rho_0 g}\,q, \qquad L_p^{-1} \ll q \ll \frac{\rho_0 g}{\overline{D}}. \qquad (5)$$

A separated scaling band requires $\rho_0 g L_p/\overline{D} \gg 1$ within the continuum regime. For $qL_p \ll 1$, the kernel tends to L_p, and Eq. (4) approaches the nonzero plateau $\overline{D}/(\overline{D} + \rho_0 g L_p)$.